

2015 9647 H2 Chemistry

Paper 1 Answer Key:

1	B	11	D	21	B	31	D
2	B	12	C	22	D	32	A
3	C	13	D	23	B	33	C
4	C	14	B	24	A	34	A
5	B	15	D	25	D	35	B
6	D	16	D	26	D	36	A
7	B	17	A	27	D	37	C
8	C	18	C	28	C	38	C
9	A	19	B	29	B	39	A
10	B	20	A	30	A	40	B

Paper 1 Worked Solutions

Q1 Answer: B

Option A does not correspond to ground state.

Option B symbolise $1s^2 2s^2 2p^2$ which clearly indicates it's a Group IV element in its ground state.

Option C is an excited state where one electron has "jumped" into the p orbital.

Option D: $1s^2 2s^2 2p^2 3s^2 3p^4$ which represent Group VI rather than Group IV.

Q2 Answer: B

σ bonds can be formed via head-on overlap between s orbital or p orbitals

π bonds cannot be established by s orbital but is possible with p orbitals

Q3 Answer: C

	electrons	neutrons
^{16}O	8	8
^1H	1	0

Since OH^- has a negative charge, there is an extra electron to consider

Total electron: 10

Total neutron: 8

Q4 Answer: C

Calcium and sodium both have giant metallic structure. Definition of metallic bonding is electrostatic forces of attraction between the cations and sea of delocalised electrons

Option A is wrong: Avoid comparison between atoms, ions will be a better choice of words.

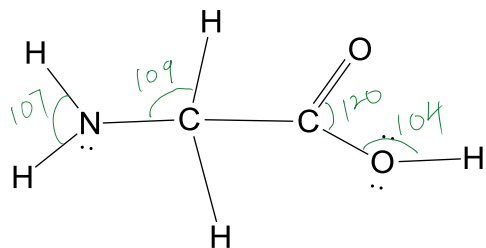
Option B wrong. Ar has no significant in melting point.

Option C: Charge density of Ca^{2+} is greater than Na^+ . This will cause the attraction between the sea of delocalised electrons and the cations to be more significant.

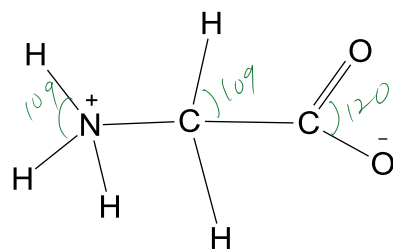
Option D: A great distractor. No doubt Ca^{2+} ion contains more electrons than Na^+ ion, in discussing metallic bonding, we tend to focus on the delocalised electrons which could be donated into the sea of delocalised electrons.

Q5 Answer: B

Aminoethanoic acid



Its zwitterion



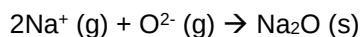
Option A: 90° bond angle is missing from the two structures

Option B: 107° is present in aminoethanoic acid but not in its zwitterion

Option C & D: 109° and 120° are present in both

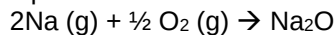
Q6 Answer: D

Lattice energy of sodium oxide:



Option A and B are wrong as the number of mole of Na_2O formed must be one mole.

Option C



represents the enthalpy change of formation rather than lattice energy.

Q7 Answer: B

$$\text{Amt of NaOH} = (12.5/1000) \times 0.05 = 0.000625 \text{ mol}$$

$$\text{Amt of HCl initially} = (25/1000) \times 0.100 = 0.0025 \text{ mol}$$

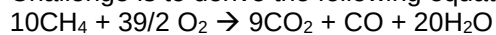
$$\begin{aligned}\text{Amt of remaining HCl} &= 0.0025 - 0.000625 \\ &= 0.001875 \text{ mol}\end{aligned}$$

$$\text{Total volume} = 12.5 + 25 = 37.5 \text{ cm}^3$$

$$[\text{HCl}] \text{ remaining} = 0.001875 / (37.5/1000) = 0.05 \text{ mol dm}^{-3}$$

Q8 Answer: C

Challenge is to derive the following equation:



[tips:

- start with $\text{CH}_4 + \text{O}_2 \rightarrow \text{CO}_2 + \text{CO} + \text{H}_2\text{O}$
- Since question mentioned mole ratio of product of CO_2 and CO , the equation will become $\text{CH}_4 + \text{O}_2 \rightarrow 9\text{CO}_2 + \text{CO} + \text{H}_2\text{O}$
- Balance the carbon, $10\text{CH}_4 + \text{O}_2 \rightarrow 9\text{CO}_2 + \text{CO} + \text{H}_2\text{O}$
- Balance the hydrogen, $10\text{CH}_4 + \text{O}_2 \rightarrow 9\text{CO}_2 + \text{CO} + 20\text{H}_2\text{O}$
- Last but not least balance the oxygen $10\text{CH}_4 + 39/2 \text{O}_2 \rightarrow 9\text{CO}_2 + \text{CO} + 20\text{H}_2\text{O}$]

Mole ratio of gases = volume ratio of gases

Since 1 dm³ of CH_4 is burned

Based on mole ratio

$$\text{Volume of O}_2 \text{ required} = 0.1 \times 39/2 = 1.95 \text{ dm}^3$$

Q9 Answer: A

If CO_2 has the ability to form hydrogen bonds with several water molecules, it will result in a more orderly configuration which will cause the ΔS to become negative.

Since bonds are formed, heat is evolved and thus forward reaction is exothermic.

Q10 Answer: B

All reagents concentration must be 1 mol dm⁻³

Q11 Answer: D

	CH ₃ CO ₂ H	CH ₃ CO ₂ ⁻	H ⁺
Initial	C	0	0
Change	-Cα	+Cα	+Cα
Eqm	C-Cα = C(1-α)	Cα	Cα

$$K_a = \frac{[H^+][CH_3CO_2^-]}{[CH_3CO_2H]}$$

$$= \frac{(C\alpha)^2}{C(1-\alpha)}$$

$$= \frac{C\alpha^2}{1-\alpha}$$

Q13 Answer: D

	H ₂ O	CO	H ₂	CO ₂
Initial	1	1	0	0
Change	-x	-x	+x	+x
Eqm	1-x	1-x	x	x

$$\text{Total mole at eqm} = 2(1-x) + 2x = 2 \text{ mol}$$

Since 33.3% of resulting mixture is hydrogen.
Amt of H₂ at eqm = 0.333 x 2 = 0.666

Let V be the volume of the mixture

$$K_c = \frac{[H_2][CO_2]}{[H_2O][CO]} = \frac{\left(\frac{0.666}{V}\right)^2}{\left(\frac{1-0.666}{V}\right)^2} = 4.0$$

Q12 Answer: C

Forward reaction is exothermic

From equation, reactants has 3 mole, product has 1 mole. Increase in pressure and decrease in temperature will result in an increase in methanol at equilibrium as equilibrium position will shift right.

Q14 Answer: B

Using the first row:

$$\text{Rate} = k[H_2][NO]^2$$

$$6 = k(0.5)(2)^2$$

$$k = 3$$

To find x x = 3(0.5)(1) ² x = 1.5	To find y y = 3(1)(1) ² y = 3.0	To find z 0.75 = 3(1)(z) ² z = 0.5
--	--	---

Q15 Answer: D

Option A is wrong: Since charge density $\propto \frac{q^+}{r^+}$
Since r⁺ increases down the group charge density will decrease

Option B is wrong: Electronegativity decreases down the group as the atomic radius increases down the group

Option C is wrong: Down Group II, the number of electron is the same.

Option D is correct: Thermal stability of Group II nitrates/carbonates increase down the Group as the charge density of the cations decrease. This means the polarising power of the cations decrease and they are less able to polarise the nitrate electron cloud effectively resulting in greater thermal stability.

Q16 Answer: D

Option A is wrong: Sulfur has two acidic oxides as well (SO₂ and SO₃).

Option B is wrong: Ar has the highest first IE (please refer to Data Booklet)

Option C is wrong: SiCl₄ can also react with water to form acidic solutions.

Option D is correct: phosphorus exists as simple molecule P₄

Q17 Answer: A

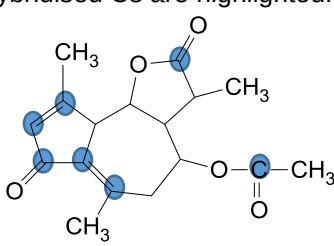
Option	Correct Description
A	Dark blue solution
B	Dark Blue solution
C	Blue ppt
D	Colourless Cu ⁺ does not allow d-d transition to occur its d orbital is completely filled.

Q18 Answer: C

Option A is wrong as Si melting point should be the highest

Option B is wrong as Al melting point should be higher than Mg

Option D is wrong as the atomic radii is of the following order: P < Si < Al < Mg

Q19 Answer: B From the write up, $\text{Cr}_2\text{O}_7^{2-}$ has been reduced to Cr^{3+}. As such $\text{Cr}_2\text{O}_7^{2-}$ is functioning as an oxidising agent. Option A is wrong: Ketone is resistant to oxidation Option B is correct: Alcohol can be oxidised by $\text{Cr}_2\text{O}_7^{2-}$ in the presence of H_2SO_4 Option C is wrong: Carboxylic acid is resistant to oxidation Option D is wrong: Under alkaline condition, $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ will not be formed as the hydroxide $\text{Cr}(\text{OH})_3$ (which is a green ppt) will be formed instead	Q20 Answer: A PCl_5 react with alcohol and carboxylic acid via nucleophilic substitution Option B is wrong as the $-\text{Br}$ cannot be substituted out. Option C is wrong as the $-\text{Cl}$ cannot be substituted out and $-\text{OH}$ should react with PCl_5 Option D is wrong as the $-\text{OH}$ should react with PCl_5										
Q21 Answer: B From the general formula, a 5 C alkyne will have the molecular formula C_5H_8. Hence, 3 structural isomers are possible. $\text{HC}\equiv\text{CCH}_2\text{CH}_2\text{CH}_3$ $\text{HC}\equiv\text{CCH}(\text{CH}_3)\text{CH}_3$ $\text{CH}_3\text{C}\equiv\text{CCH}_2\text{CH}_3$ Remember that there are two ways to arrange $-\text{C}_3\text{H}_7$	Q22 Answer: D The sp^2 hybridised Cs are highlighted. 										
Q23 Answer: B Alkenes will not be reduced by NaBH_4. Aldehydes will be reduced to primary alcohols. Hence, product formed is $\text{CH}_2\text{CHCH}_2\text{CH}_2\text{OH}$.	Q24 Answer: A Citric acid has the molecular formula $\text{C}_6\text{H}_8\text{O}_7$. Observing the information given in the question (ie. conc H_2SO_4, and only carboxylic acids and alcohols are given in the options), the reaction that occurred is esterification. Hence, either the alcohol or carboxylic acid in citric acid has reacted. If the $-\text{COOH}$ groups in citric acid have reacted to form ester, there will not be any increase in number of O. If the $-\text{OH}$ group in citric acid has reacted to form an ester, there will be an increase in 1 O. Since an organic product of $\text{C}_9\text{H}_{12}\text{O}_8$ is formed, with an increase in 3 C and 1 O, the $-\text{OH}$ group in citric acid has reacted with a 3 C carboxylic acid. Draw the product out and check that the number of H is correct.										
Q25 Answer: D When an organic compound reacts with AgNO_3 (aq), hydrolysis has taken place. The nucleophile is H_2O. Ease of hydrolysis: <table style="width: 100%; text-align: center;"><tr><td>RCOCl</td><td>$>$</td><td>RCI</td><td>$>$</td><td>ArCl</td></tr><tr><td>acyl chloride</td><td></td><td>halogenoalkane</td><td></td><td>halogenoarene</td></tr></table> Acyl chloride reacts readily with water to give carboxylic acid and hydrogen chloride gas. $\text{Cl}^- (\text{aq}) + \text{Ag}^+ (\text{aq}) \rightarrow \text{AgCl} (\text{s})$ Option A is incorrect as there is absence of energetically accessible vacant orbitals in C of CCl_4 to undergo hydrolysis reaction as H_2O cannot form dative bonds	RCOCl	$>$	RCI	$>$	ArCl	acyl chloride		halogenoalkane		halogenoarene	Q26 Answer: D At pH 11, the medium is alkaline. Hence, the two $-\text{COOH}$ groups in aspartic acid will react with OH^- present to form $-\text{COO}^-$.
RCOCl	$>$	RCI	$>$	ArCl							
acyl chloride		halogenoalkane		halogenoarene							

with the central C atom. Also, the 4 bulky Cl atoms in CCl₄ hinders H₂O from attacking the electron deficient C.

Option B is incorrect as the C-Cl bond in chlorobenzene has partial double character. Hence, halogenoarene does not undergo nucleophilic substitution reactions.

Q27 Answer: D

Option	Q	R
A	RCHO	RCOOH
	Increase in M _r = 16.0	
B	Aldehydes are not oxidised to ketones	
C	RCH ₂ OH	RCHO
	Decrease in M _r = 2.0	
D	RCH ₂ OH	RCOOH
	Increase in M _r = 14.0	

Q28 Answer: C

Both phenol and alcohol are present in the compound. Phenols, being acidic, will react with NaOH (aq) to form phenoxide. Alcohols, being neutral, will not react with NaOH (aq).

Q29 Answer: B

Electrophilic substitution has taken place. The hydroxyl group (which is 2,4-directing) strongly activates the benzene ring towards easy electrophilic substitution. Halogenation occurs at room temperature without halogen carrier.

Using Br₂ (aq) will result in multiple substitutions.

Q30 Answer: A

An electrophilic addition reaction is shown. The intermediate is a carbocation with the Br group in it as the electrophile is the δ⁺ Br.

Q31 Answer: D

The 3d electronic configuration of these transition metal ions are as follows:

Mn ²⁺	3d ⁵ (no paired 3d electrons as all orbitals are singly filled)
Fe ²⁺	3d ⁶ (one electron pair in one of the 3d orbitals)
Co ³⁺	3d ⁶ (one electron pair in one of the 3d orbitals)

Q32 Answer: A

All 3 compounds are covalent.

Both (CH₃)₂SiCl₂ and GeCl₄ have low boiling point. In AlBr₃, Al³⁺ has high charge density and Br⁻ has a large electron cloud. Thus, there is significant polarisation of the Br⁻ electron cloud by Al³⁺. AlBr₃ is a simple molecular compound.

Q33 Answer: C

The 4 assumptions of the kinetic theory as applied to an ideal gas:

1. The particles (atoms or molecules) of a gas are in constant and random motion.
2. The gas particles have negligible volume (or size) when compared to that of the container.
3. The intermolecular forces of attraction between the gas particles as well as between the gas particles and walls of the container are negligible.
4. The collisions between gas particles are perfectly elastic, bouncing apart on collision with no loss in kinetic energy.

Molecules have mass. Hence, statement 1 is incorrect.

Q34 Answer: A

Reduction occurs at the carbon terminal (more positive reduction potential of +0.74V) and oxidation occurs at the zinc terminal. NH₄⁺ is reduced to NH₃ and H₂ and Zn is oxidised to Zn²⁺. Hence, Zn is used up and the zinc casing becomes thinner.

$$\begin{aligned}
 E^{\ominus}_{\text{cell}} &= E^{\ominus}_{\text{red}} - E^{\ominus}_{\text{oxid}} \\
 &= +0.74 - (-0.76) \\
 &= +1.50 \text{ V}
 \end{aligned}$$

Equation for cell reaction:
 $2\text{NH}_4^+ (\text{aq}) + \text{Zn} (\text{s}) \rightarrow \text{Zn}^{2+} (\text{aq}) + 2\text{NH}_3 (\text{g}) + \text{H}_2 (\text{g})$

Q35 Answer: B

$$k = A e^{-E_a / RT}$$

Rate constant, k , is affected by the presence of a catalyst as well as changes in temperature.

Adding a catalyst will lower the E_a of both the forward and backward reaction by the same extent. Hence, both k_f and k_b will increase by the same extent.

Increasing the temperature will increase both k_f and k_b . However, as the forward reaction is an exothermic reaction, and increasing temperature will favour the endothermic reaction, k_f will increase less than k_b .

Q37 Answer: C

A free radical always has an unpaired electron and can be formed from homolytic fission ($\text{Cl}_2 \rightarrow 2\text{Cl}\bullet$).

It need not have a lone pair (eg. $\text{H}\bullet$).

Q36 Answer: A

Statement 1 is correct.

In methanal, HCHO , the carbon is sp^2 hybridised and the whole molecule is planar.

Statement 2 is correct.

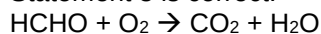
Oxidation no. of O: -2

Oxidation no. of H: +1

Whole molecule is neutral.

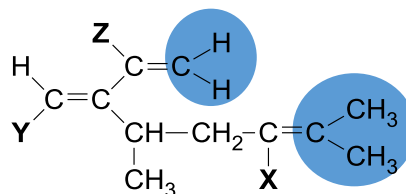
Hence, oxidation no. of C is 0.

Statement 3 is correct.



Q38 Answer: C

Regardless of the identities of **Z** and **X**, these two atoms will not result in geometrical (or cis-trans) isomerism. This is because the $2=\text{C}$ highlighted are bonded to the same groups.



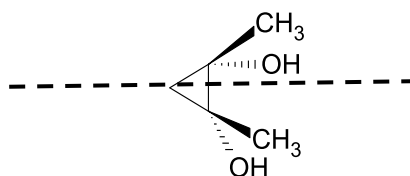
Y cannot be H atom. If not, the entire compound will not exhibit geometrical isomerism.

Q39 Answer: A

Thymol contains a phenol and benzylic H. Hence, it can react with ethanoyl chloride (nucleophilic substitution) and sodium metal (redox). With the presence of benzylic H, thymol can undergo oxidation with alkaline KMnO_4 (aq).

Q40 Answer: B

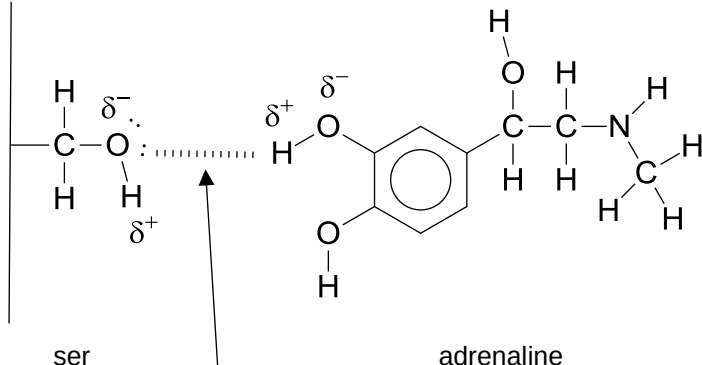
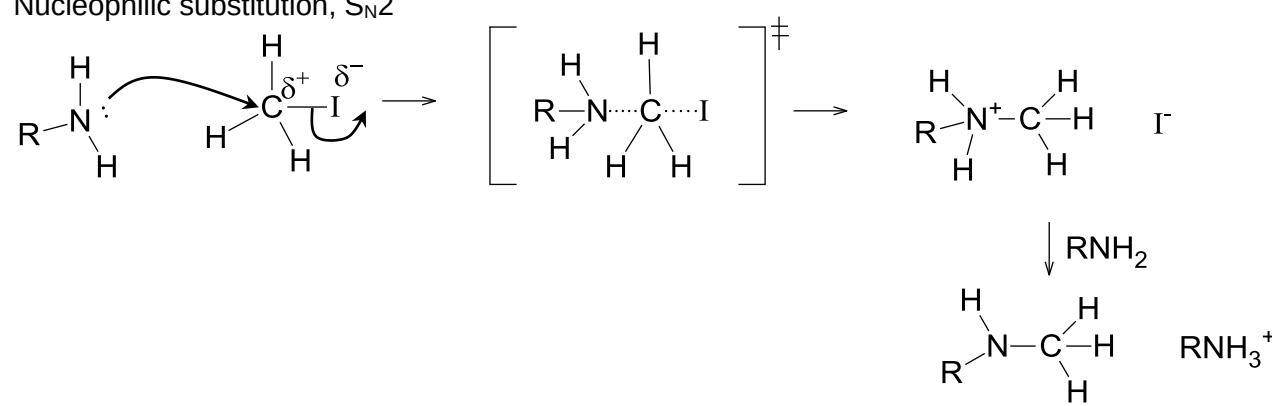
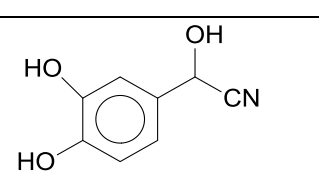
Molecules with a plane of symmetry will not be optically active. Only the molecule in option 3 has a plane of symmetry.

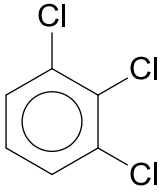
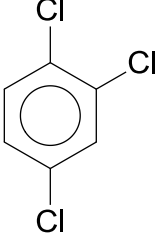
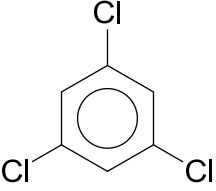
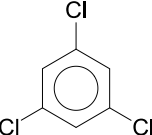
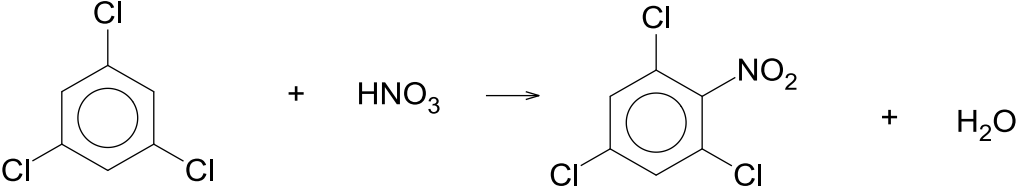


Paper 2 Worked Solutions

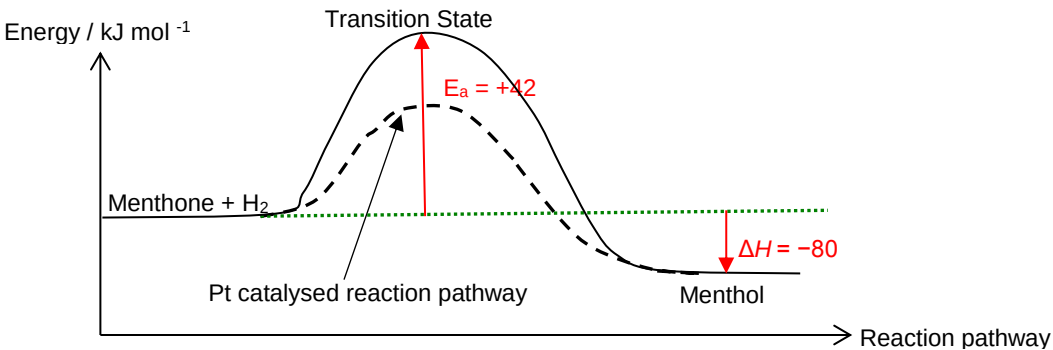
2(a)	(i)	Standard enthalpy change of combustion ($\Delta H_c^\ominus$) of a substance is the energy evolved when one mole of the substance is completely burnt in oxygen under standard conditions .
2(a)	(ii)	Petrol consists of a mixture of hydrocarbons. $\Delta H_c^\ominus$ cannot be determined.
2(b)	(i)	Using $pV = nRT$ At 298K, 24.8×10^6 Pa and $1 \times 10^{-3} \text{ m}^3$, (note do convert 1 dm^3 to m^3) $n = \frac{24.8 \times 10^6 \times 1 \times 10^{-3}}{8.31 \times 298} = 10.0 \text{ mol}$ mass of methane = $10.0 \times (12+4) = 160\text{g}$ energy released = $160 \times 55.5 = \mathbf{8880 \text{ kJ}}$ Alternative method: $101 \text{ kPa} \rightarrow 0.0358 \text{ MJ dm}^{-3}$ $24.8 \text{ MPa} \rightarrow (24.8 \times 0.0358 \times 10^6) \div 101 \times 10^3 = 8790$
2(b)	(ii)	The combustion of hydrogen produces water, which is not harmful to the environment. Methane, on the other hand, combusts to produce carbon dioxide which is a greenhouse gas.
2(c)		Both methanol and methane have simple molecular structures. Methanol has stronger intermolecular hydrogen bonding between molecules as compared to the weaker intermolecular instantaneous dipole-induced dipole interactions between methane molecules. Thus, more energy is required to overcome these stronger intermolecular forces of attraction of methanol than methane. There is also more electrons present in methanol and thus the probability of unsymmetrical distribution of electrons increases and as such there will be stronger Van der Waals' forces of attraction for methanol than methane.
2(d)	(i)	$\Delta H^\ominus = \Delta H_c^\ominus (\text{reactants}) - \Delta H_c^\ominus (\text{products})$ $= -726 - (-2 \times 286 + (-283)) \quad (\text{don't forget to multiple by 2})$ $= \mathbf{+129 \text{ kJ mol}^{-1}}$
2(d)	(ii)	For reaction to be spontaneous, $\Delta G < 0$ $\Delta G^\ominus = \Delta H^\ominus - T\Delta S^\ominus$ $0 > 129 - T(+0.332)$ $T > \mathbf{389K}$
2(e)		CO is a stronger ligand than O_2 and bonds irreversibly via a stronger dative bond with the iron atom of haemoglobin, thus preventing haemoglobin from carrying oxygen to be transported around the body.

3(a)(i)	$\Delta H = \sum BE(\text{rxts}) - \sum BE(\text{pdts}) = (436 + 244) - (2 \times 431)$ $= \underline{-182 \text{ kJ mol}^{-1}}$
3(a)(ii)	*Show state symbols clearly, arrow head must touch the lines.
3(b)(i)	<u>Light provides the energy required</u> to break the Cl-Cl covalent bond via homolytic fission <u>to form</u> Cl free <u>radicals</u> .
3(b)(ii)	<u>Free radical substitution</u>
3(b)(iii)	After initiation step, <u>more radicals are produced and Cl radicals are regenerated</u> , resulting in a <u>chain reaction</u> . The production of new radicals is an <u>exothermic reaction</u> , causing an increase in reaction rate.
3(c)	$\Delta H_f \text{ HBr} = \sum BE(\text{rxts}) - \sum BE(\text{pdts}) = (436 + 193) - (2 \times 366) = -103 \text{ kJ mol}^{-1}$ $\Delta H_f \text{ HCl} = -182 \text{ kJ mol}^{-1} \text{ (from 3(a)(i))}$ Since <u>BE of H-Br (366 kJ mol⁻¹) is lower than BE of H-Cl (431 kJ mol⁻¹)</u>, <u>less energy is given out</u> during the overall reaction involving bromine. The overall <u>ΔH</u> involving bromine is <u>less exothermic</u> as compared to that with chlorine.
3(d)(i)	$\text{NaCl} + \text{H}_2\text{SO}_4 \rightarrow \text{HCl} + \text{NaHSO}_4$
3(d)(ii)	$\text{NaBr} + \text{H}_2\text{SO}_4 \rightarrow \text{HBr} + \text{NaHSO}_4$ $2\text{HBr} + \text{H}_2\text{SO}_4 \rightarrow \text{SO}_2 + \text{Br}_2 + 2\text{H}_2\text{O}$ The HBr produced in the first step will react with more H ₂ SO ₄ to release Br ₂ as <u>Br⁻ is a stronger reducing agent than Cl⁻</u> , as it is able to reduce H ₂ SO ₄ to SO ₂ .
3(e)	Test: Bubble chlorine gas or Cl ₂ (aq) into both solutions separately Observations: <u>No colour change</u> / Solution remains <u>colourless</u> for HCl, but <u>reddish brown Br₂</u> upon reaction with HBr Equation (not required): $\text{Cl}_2 + 2\text{Br}^- \rightarrow \text{Br}_2 + 2\text{Cl}^-$

4(a)		 ser adrenaline hydrogen bond *please draw displayed formula
4(b)	(i)	Nucleophilic substitution, S_N2 
4(b)	(ii)	Primary carbocation will be unstable and thus will not favour S_N1 mechanism. Moreover, primary halogenoalkane shows little steric hindrance, as such, nucleophile can approach the partial positive charged carbon easily, thus allow S_N2 to proceed.
4(c)	(i)	 Step 1: HCN with trace amount of NaCN, 10 °C to 20 °C or cold. Step 2: LiAlH₄ in dry ether, r.t.p.
4(c)	(ii)	Presence of racemic mixture. There is equal probability for the CN⁻ nucleophile to attack from the top or the bottom of the trigonal planar aldehyde functional group in 3,4-dihydroxybenzaldehyde. Since one optical isomer of noradrenaline can bind to the protein receptor via hydrogen bonding, 50% of the drug synthesised is active in human body.
4(d)	(i)	$\text{Ag}_2\text{O} + 2 \text{H}^+ + 2 \text{e}^- \rightarrow 2\text{Ag} + \text{H}_2\text{O}$
4(d)	(ii)	Reduction : $\text{Ag}_2\text{O} + 2 \text{H}^+ + 2 \text{e}^- \rightarrow 2\text{Ag} + \text{H}_2\text{O}$ ----(1) Oxidation: $\text{C}_9\text{H}_{13}\text{NO}_3 \rightarrow \text{C}_9\text{H}_9\text{NO}_3 + 4\text{H}^+ + 4\text{e}^-$ ----(2) (1) $\times$ 2 + (2) Overall : $2\text{Ag}_2\text{O} + \text{C}_9\text{H}_{13}\text{NO}_3 \rightarrow 4\text{Ag} + \text{C}_9\text{H}_9\text{NO}_3 + 2 \text{H}_2\text{O}$ Amt of adrenaline in 50 cm³ = $(3.0 \times 10^{-3}) \times 50/1000$ = 1.50×10^{-4} mol $2\text{Ag}_2\text{O} \equiv \text{C}_9\text{H}_{13}\text{NO}_3$ Amit of Ag₂O = 3.00×10^{-4} mol Mass of Ag₂O = $3 \times 10^{-4} \times [2(108) + 16]$ = 0.0696 g

5(a)	Compound X and Y do not undergo nucleophilic substitution reaction ⇒ They are halogenoarene, ⇒ Strengthening of the C-Cl bond due to formation of partial double bond character in C-X bond The p orbital of the halogen atom can overlap with the π-electron cloud of the benzene ring. As a result, the lone pair of electrons on the halogen atom can delocalise into the benzene ring.																						
5(b)	(i)	<table border="1"><thead><tr><th></th><th>C</th><th>H</th><th>Cl</th></tr></thead><tbody><tr><td>Percentage</td><td>49</td><td>2.7</td><td>48.3</td></tr><tr><td>Ar</td><td>12</td><td>1</td><td>35.5</td></tr><tr><td>Amt/mol</td><td>4.083</td><td>2.7</td><td>1.361</td></tr><tr><td>Simplest ratio</td><td>3</td><td>2</td><td>1</td></tr></tbody></table> Empirical Formula: C₃H₂Cl			C	H	Cl	Percentage	49	2.7	48.3	Ar	12	1	35.5	Amt/mol	4.083	2.7	1.361	Simplest ratio	3	2	1
	C	H	Cl																				
Percentage	49	2.7	48.3																				
Ar	12	1	35.5																				
Amt/mol	4.083	2.7	1.361																				
Simplest ratio	3	2	1																				
5(b)	(ii)	pV = nRT 101000 × 87.4 × 10⁻⁶ = 0.344 / Mr × 8.31 × 454 Mr = 147.0 Let the molecular formula of X be (C₃H₂Cl)_n n (3 × 12 + 2 × 1.0 + 35.5) = 147 Molecular formula of X is C₆H₄Cl₂																					
5(c)	 1,2,3-trichlorobenzene  1, 2, 4- trichlorobenzene  1, 3,5- trichlorobenzene																						
5(d)	Compound Y is:  																						

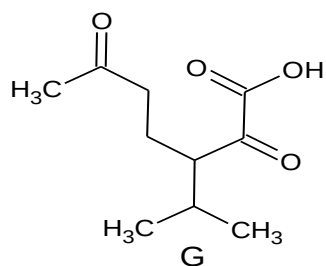
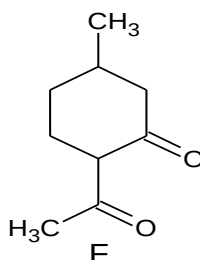
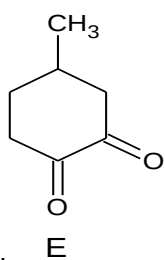
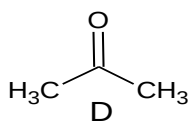
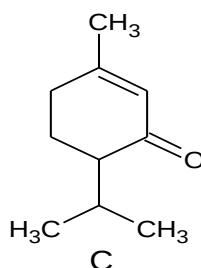
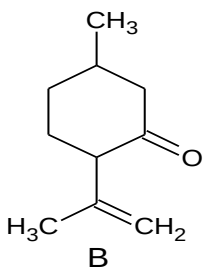
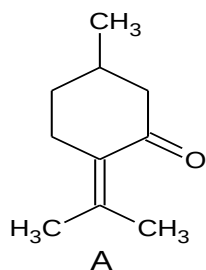
Paper 3 Worked Solutions

1	(a) (i)	$\text{R-OH} + \text{Na} \rightarrow \text{R-O}^-\text{Na}^+ + \frac{1}{2}\text{H}_2$ Amt of menthol present = $2 \times 1.32 \times 10^{-2} \text{ mol}$ Mass of menthol present = $2 \times 1.32 \times 10^{-2} \times 156 = 4.118 \text{ g}$ % by mass = $4.118 \div 10 \times 100 = 41.2\%$	(ii)	No. of chiral carbons = 3 No. of optical isomers possible = $2^3 = 8$
	(b) (i) (ii)			
	(iii)	Catalysts provide an alternative reaction pathway of a lower activation energy which increases the rate of reaction. The H ₂ and menthone molecules are adsorbed on the surface of Ni metal. This weakens both H-H σ bonds and C=O π bonds, and the formation of C-H and O-H bonds take place readily. After menthol is formed, it desorbs from the surface of Ni metal.		
	(iv)	Palladium Palladium has the same outermost electronic configuration as nickel and platinum, all of which belongs to the same group. Hence it would allow for adsorption of H ₂ and methone molecules.		
	(c)	A, B and C have molecular formula C₁₀H₁₆O - Number of C and H atoms are not comparable - Hence benzene is not present A, B and C decolourises Br₂ (aq) - Electrophilic addition occurred - A, B and C contain alkene functional groups A, B and C all produces orange ppt with 2,4-DNPH - Condensation reaction occurred - A, B and C contain carbonyl functional groups A, B and C does not reduce Fehling's solution - A, B and C are not oxidised by Fehling's solution - A, B and C contain ketone functional groups A, B and C forms menthol after reaction by H₂ and Pt catalyst - Reduction occurred (C₁₀H₁₆O to C₁₀H₂₀O) - Ketone functional group reduced - Alkene functional group reduced - Since only 4 atoms of H were added hence there is only 1 alkene group present in A, B and C A produces two compounds D and E when reacted with hot KMnO₄ - Vigorous oxidation of alkene occurred - Alkene functional group is not terminal B produces F and CO₂ when reacted with hot KMnO₄ - Vigorous oxidation of alkene occurred - F has one less carbon than B - Alkene functional group in B is terminal C produces G when reacted with hot KMnO₄ - Vigorous oxidation of alkene occurred		

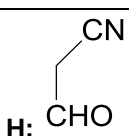
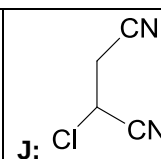
- G has the same number of carbons as C, and has three more oxygen atoms than C
- Double bond is broken to form a COOH group and a ketone group

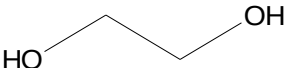
D, F and G produces yellow ppt with I_2 (aq)

- Oxidation occurred
- Presence of carbonyl or alcohol terminal methyl functional groups
- Since D, F and G are products of $KMnO_4$ oxidation, this indicates only keto terminal methyl functional group



2(a)	Volatility of the halogens <u>decreases</u> down the group. Halogens have <u>simple molecular structures</u> and are <u>non-polar</u> . Down the group, the <u>number of electrons</u> in the halogen molecules <u>increases</u> . There is <u>greater unsymmetrical distribution of the electron cloud</u> . <u>Intermolecular instantaneous dipole-induced dipole interactions</u> become <u>more extensive</u> and require <u>more energy</u> to overcome.													
2(b)	$6\text{NaOH}(\text{aq}) + 3\text{Cl}_2(\text{aq}) \rightarrow 5\text{NaCl}(\text{aq}) + \text{NaClO}_3(\text{aq}) + 3\text{H}_2\text{O}(\text{l})$ The oxidation number of Cl increases from <u>0 in Cl_2</u> to <u>+5 in NaClO_3</u> and decreases from <u>0 in Cl_2</u> to <u>-1 in NaCl</u> .													
2(c)	(i)	$\text{BE}(\text{H-F}) = 562 \text{ kJmol}^{-1}$, $\text{BE}(\text{H-Cl}) = 431 \text{ kJmol}^{-1}$, $\text{BE}(\text{H-Br}) = 366 \text{ kJmol}^{-1}$, $\text{BE}(\text{H-I}) = 299 \text{ kJmol}^{-1}$ The <u>bond energy of HF is the highest</u> among the halogen halides. This implies that the <u>bond strength of HF is greatest</u> compared to other halogen halides, hence <u>more energy</u> is required to <u>break the covalent bonds in H-F</u> for it to dissociate. Hence HF is a weak acid which <u>dissociates partially</u> .												
	(ii)	$[\text{HCl}] = [\text{H}^+] = 0.5 \text{ moldm}^{-3}$ $\text{pH of HCl} = -\lg[\text{H}^+] = -\lg(0.5) = \underline{0.301}$ $[\text{H}^+] \text{ in a weak acid, HF,} = \sqrt{K_a \times c} = \sqrt{5.6 \times 10^{-4} \times 0.5} = 0.0167 \text{ moldm}^{-3}$ $\text{pH of HF} = -\lg[\text{H}^+] = -\lg(0.0167) = \underline{1.77}$												
2(d)	$\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(\text{s})$ <u>White ppt of AgCl</u> is formed. $\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightleftharpoons \text{AgCl}(\text{s})$ $\text{Ag}^+(\text{aq}) + 2\text{NH}_3(\text{aq}) \rightleftharpoons [\text{Ag}(\text{NH}_3)_2]^+(\text{aq})$ When $\text{NH}_3(\text{aq})$ is added, AgCl dissolves to form a <u>colourless solution</u> containing the complex, <u>$[\text{Ag}(\text{NH}_3)_2]^+\text{Cl}^-$</u> . $\text{Ag}^+(\text{aq}) + \text{I}^-(\text{aq}) \rightarrow \text{AgI}(\text{s})$ <u>Yellow ppt of AgI</u> is formed. <u>AgI does not dissolve and a yellow precipitate remains</u> when $\text{NH}_3(\text{aq})$ is added.													
2(e)	(i)	The value of pV should remain <u>constant</u> at all values of p. (applying $\text{PV} = nRT$, PV = constant at constant T)												
	(ii)	$\text{pV} = nRT$ $V = \frac{nRT}{P} = \frac{0.4 \times 8.31 \times 300}{12 \times 10^{-5}} = \underline{8.31 \times 10^{-4} \text{ m}^3}$ $V = \underline{0.831 \text{ dm}^3}$												
	(iii)	<table border="1"><thead><tr><th>Pressure, p/Pa</th><th>Volume, V/dm³</th><th>pV/ Pa dm³</th></tr></thead><tbody><tr><td>5.0×10^5</td><td>1.924</td><td><u>9.62×10^5</u></td></tr><tr><td>10.0×10^5</td><td>0.926</td><td><u>9.26×10^5</u></td></tr><tr><td>15.0×10^5</td><td>0.592</td><td><u>8.88×10^5</u></td></tr></tbody></table> Taking proportion: increase of 1×10^5 in pressure, volume decrease by $(9.25 \times 10^5 - 8.88 \times 10^5) \div 5 = 7400$ Thus with pressure of $12.0 \times 10^5 = 9.26 \times 10^5 - 7400 - 7400 = 911200 \text{ Pa dm}^3$ When pV = 911200, $V = 911200 \div (12 \times 10^5) = 0.759 \text{ dm}^3$	Pressure, p/Pa	Volume, V/dm ³	pV/ Pa dm ³	5.0×10^5	1.924	<u>9.62×10^5</u>	10.0×10^5	0.926	<u>9.26×10^5</u>	15.0×10^5	0.592	<u>8.88×10^5</u>
Pressure, p/Pa	Volume, V/dm ³	pV/ Pa dm ³												
5.0×10^5	1.924	<u>9.62×10^5</u>												
10.0×10^5	0.926	<u>9.26×10^5</u>												
15.0×10^5	0.592	<u>8.88×10^5</u>												
	(iv)	Volume calculated in (iii) is smaller. HCl is not an ideal gas because it has <u>significant intermolecular forces of attraction</u> which are not negligible. As HCl is a <u>polar molecule</u> , there is <u>permanent-dipole permanent-dipole (pd-pd) interaction</u> between HCl molecules and <u>deviates from ideal gas behaviour</u> . Therefore it occupies a smaller volume.												
2(f)	(i)	$\Delta G = \Delta H - T\Delta S$ $0 = 16.8 \times 10^3 - (188)(\Delta S)$ $\Delta S = \underline{+0.0894 \text{ kJ mol}^{-1}\text{K}^{-1} \text{ or } +89.4 \text{ J mol}^{-1}\text{K}^{-1}}$ The sign is <u>positive</u> because there is an <u>increase in change in entropy</u> from a <u>phase change of HCl from liquid to gas</u> , which results in an increase in the number of gaseous molecules (hence disorder) There are more ways to distribute the energy.												
	(ii)	$\Delta G = \Delta H - T\Delta S$ $\Delta G = 16.8 - (298)(0.0894) = \underline{-9.83 \text{ kJ mol}^{-1}}$ The sign is <u>negative</u> . It implies that the vapourisation of HCl is a <u>spontaneous</u> process at 298 K.												

3(a)(i)	Magnesium burns vigorously/ very readily with oxygen with a bright white flame producing white magnesium oxide solid/ ash , while calcium burns vigorously/ very readily with oxygen with a brick-red flame producing white calcium oxide solid .
3(a)(ii)	MgO has low solubility in water (due to high lattice energy). A small amount of Mg(OH) ₂ is formed. CaO dissolves readily in water to produce a strongly alkaline solution , Ca(OH) ₂ , with pH 13.
3(b)(i)	0.2 mol dm ⁻³ (or any suitable value between solubility: 2.5 x 10 ⁻² mol dm ⁻³ - Ca(OH) ₂ and 4.1 x 10 ⁻¹ mol dm ⁻³ - Ba(OH) ₂)
3(b)(ii)	Ca(OH) ₂ ⇌ Ca ²⁺ + 2OH ⁻ [OH ⁻] = 2 x (2.5 x 10 ⁻²) = 5.0 x 10 ⁻² mol dm ⁻³ pH = 14 - pOH = 14 - (-lg(5.0 x 10 ⁻²)) = 12.7
3(b)(iii)	K _{sp} = [Mg ²⁺][OH ⁻] ² K _{sp} = (1.6 x 10 ⁻⁴)(2 x 1.6 x 10 ⁻⁴) ² = 1.64 x 10⁻¹¹ mol³ dm⁻⁹
3(b)(iv)	A white precipitate is observed. Common ion effect due to the presence of OH⁻ from both Mg(OH) ₂ and Ba(OH) ₂ will cause equilibrium position of both hydroxides M(OH)₂ ⇌ M²⁺ + 2OH⁻ to shift to the left forming M(OH) ₂ , decreasing their solubilities. Ionic product of Mg(OH) ₂ = $\left(\frac{1.6 \times 10^{-4}}{2}\right) \left(\frac{(2 \times 1.6 \times 10^{-4}) + 2(4.1 \times 10^{-1})}{2}\right)^2 = 1.35 \times 10^{-5} \text{ mol}^3 \text{ dm}^{-9}$. Since IP > K_{sp} of Mg(OH)₂ , precipitation of Mg(OH) ₂ occurs. Ionic product of Ba(OH) ₂ = $\left(\frac{4.1 \times 10^{-1}}{2}\right) \left(\frac{(2 \times 1.6 \times 10^{-4}) + 2(4.1 \times 10^{-1})}{2}\right)^2 = 3.44 \times 10^{-2} \text{ mol}^3 \text{ dm}^{-9}$. Since IP < K_{sp} of Ba(OH)₂ (2.76 x 10 ⁻¹ mol ³ dm ⁻⁹), precipitation of Ba(OH) ₂ does not occur. *Note [] are halved as equal volumes of Mg(OH) ₂ and Ba(OH) ₂ are mixed
3(c)(i)	Amino acids: asp and glu They are able to form ionic bonds with Ca ²⁺ ions.
3(c)(ii)	Regular coiled configuration consisting of 3.6 amino acids per helical turn Hydrogen bonds are formed between O atom of (-C=O) group and the H atom of (-N-H) group of the fourth peptide bond down the same protein chain, which stabilises the structure. R groups on the α-carbons all points outwards and are perpendicular to the main axis of the helix
3(c)(iii)	HCl(aq) with heat (or any other strong aqueous acid or alkali)
3(c)(iv)	gly-asp-gly-tyr-ile-ser
3(d)(i)	 H: CHO  J: Cl
3(d)(ii)	Step 4: alcoholic NH ₃ , heat in sealed tube Step 5: HCl(aq) with heat (or H ₂ SO ₄ (aq) with heat)

4(a)	Step 1: Hydrolysis Step 2: Elimination Step 3: Reduction
4(b)	$\text{C}_6\text{H}_8\text{O} + 15/2 \text{O}_2 \rightarrow 6 \text{CO}_2 + 4 \text{H}_2\text{O}$ $\Delta H = \text{BE}(\text{reactants}) - \text{BE}(\text{products})$ $= [8(\text{C-H}) + 2(\text{C=C}) + 3(\text{C-C}) + 2(\text{C-O}) + 7.5(\text{O=O})] - [12(\text{C=O}) + 8(\text{O-H})]$ $= [8(410) + 2(610) + 3(350) + 2(360) + 7.5(496)] - [12(805) + 8(460)]$ $= \underline{\underline{-3350 \text{ kJ mol}^{-1}}}$
4(c)	$Q' = mc\Delta T = 200 \times 4.18 \times 32 = 26752 \text{ J}$ $Q' = 0.8Q$ $Q = 33440 \text{ J}$ $\Delta H = -\frac{Q}{n} = \frac{-33440}{1/96} = \underline{\underline{-3210 \text{ kJ mol}^{-1}}}$ The bond energies are all average values. Or Liquid DMF was combusted. That means some energy was used to convert the liquid DMF to gaseous DMF.
4(d)	(i) Mechanism: $\text{R-CH}_2\text{OH} + \text{H}^+ \xrightarrow{\text{fast}} \text{R-CH}_2\text{O}^+\text{H}_2$ $\text{R-CH}_2\text{O}^+\text{H}_2 \xrightarrow{\text{slow}} \text{R-C}^+\text{H}_2 + \text{H}_2\text{O}$ $\text{R-C}^+\text{H}_2 + \text{Cl}^- \xrightarrow{\text{fast}} \text{R-CH}_2\text{Cl}$ The second step is the rate-determining step.
4(d)	(ii) $\text{K}_2\text{Cr}_2\text{O}_7$, $\text{H}_2\text{SO}_4(\text{aq})$, heat
4(e)	(i) Ester
4(e)	(ii) 
4(e)	(iii) Concentrated H_2SO_4 , heat
4(f)	Initiation: $\text{Cl}_2 \xrightarrow{\text{UV}} 2\text{Cl}^\bullet$ Propagation: $\text{R-CH}_3 + \text{Cl}^\bullet \rightarrow \text{R-CH}_2^\bullet + \text{HCl}$ $\text{R-CH}_2^\bullet + \text{Cl}_2 \rightarrow \text{R-CH}_2\text{Cl} + \text{Cl}^\bullet$ Termination: $\text{Cl}^\bullet + \text{Cl}^\bullet \rightarrow \text{Cl}_2$ $\text{R-CH}_2^\bullet + \text{Cl}^\bullet \rightarrow \text{RCH}_2\text{Cl}$ $\text{R-CH}_2^\bullet + \text{R-CH}_2^\bullet \rightarrow \text{R-CH}_2\text{CH}_2\text{-R}$

5(a)	(i)	Proton number is the number of protons in the nucleus of an atom of an element. Nucleon number is the total number of protons and neutrons in the nucleus of an atom of an element.
5(a)	(ii)	Let the percentage abundance of ${}^6\text{Li}$ be $x\%$; ${}^7\text{Li}$ will be $(100 - x)\%$ $x(6.015) + (100 - x)(7.016) = 6.942 \times 100$ $x = 7.39$ Percentage abundance of ${}^6\text{Li}$ is 7.39% Percentage abundance of ${}^7\text{Li}$ is 92.61%
5(a)	(iii)	${}^6\text{Li} + {}^2\text{H} \rightarrow {}^4\text{He} + {}^3_2\text{He} + {}^1_0\text{n}$ (X is ${}^3_2\text{He}$) ${}^6\text{Li} + {}^2\text{H} \rightarrow {}^7_3\text{Li} + {}^1_1\text{p}$ (Y is ${}^7_3\text{Li}$)
5(b)	(i)	Metallic bonding. Interaction between the sea of delocalised electron in Li^+ ions and sea of delocalised electrons in the graphite.
	(ii)	Before discharge: Oxidation number of Co is +4 in CoO_2 After discharge: Oxidation number of Co is + 3
	(iii)	There are 4 bond pairs and 0 lone pair around B atom To minimise repulsion and maximise stability, the 4 electron pairs are directed to the corners of a regular tetrahedron BF_4^- is tetrahedral
	(iv)	KMnO_4 in dilute H_2SO_4 (or dilute NaOH), cold
	(v)	Nucleophilic substitution
(c)	(i)	$\text{Li}_2\text{O}_2 + \text{CO}_2 \rightarrow \text{Li}_2\text{CO}_3 + \frac{1}{2} \text{O}_2$ (exhaled air) (breathable air)
	(ii)	High charge density and hence high polarising power of Li^+ C–O bond in the CO_3^{2-} anion is polarised and weakened by the Li^+ such that the C–O bond breaks and Li_2O and CO_2 are formed.
(d)	(i)	$\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ and $\text{CH}_3\text{CH}_2\text{CHO}$ or $\text{CH}_3\text{CH}_2\text{Br}$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$
	(ii)	$(\text{CH}_3\text{CH}_2)_2\text{CO}$ and CH_3Br
(e)		Test: Add acidified $\text{K}_2\text{Cr}_2\text{O}_7$ (aq) to each test tube containing P, Q and R and heat . Observation: orange $\text{K}_2\text{Cr}_2\text{O}_7$ turns green for P and Q ; solution remains orange for R Test: Add 2,4-DNPH to the test tube from above containing the oxidised products and warm . Observation: orange ppt for oxidised product of Q and no orange ppt for oxidised product of P

END